Operations Research

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Exercise Sheet 8

Training Exercises

Exercise T8.1. *Matroids*

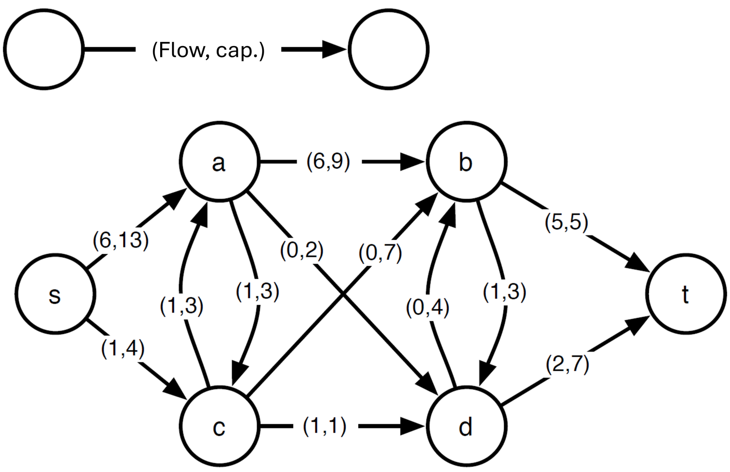
Consider two matroids $U_1 = (E_1, \mathcal{I}_1)$ and $U_2 = (E_2, \mathcal{I}_2)$.

a) Show that the family of sets $U_1 \cap U_2 := (E_1 \cap E_2, \mathcal{I}_1 \cap \mathcal{I}_2)$ is an independence system ("Unabhängigkeitssystem").

b) Refute that in general the set system $U_1 \cap U_2 := (E_1 \cap E_2, \mathcal{I}_1 \cap \mathcal{I}_2)$ is a matroid.

Exercise T8.2. *Maximum Flow*

Consider the following network with flow (from s to t) and capacities as indicated.



a) State the current flow value.

b) Based on the current flow, perform two more flow augmentations using the Ford-Fulkerson algorithm. Specify the two associated paths, as well as the resulting flow.

c) Is the obtained flow maximal? Give reasons for your answer. What is the maximum flow value here?

d) Specify a minimum s, t cut.

Exercise T8. 3. *Dredgers*

A construction company has 5 dredgers and needs one of them at each of their 4 construction sites. The distances between the dredgers and the construction sites are given as follows.

dredge	site			
	A	B	C	D
1	90	75	75	80
2	35	85	55	65
3	125	95	90	105
4	45	110	95	115
5	55	85	60	85

Minimize the total transport distance using the Hungarian method. Then specify the optimal assignment and the total transport distance.

Self-study Exercises

Exercise S8. 1. *TU and dual of a Matroid*

a) Given the matrix $A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$. Determine whether A is totally unimodular. Justify your answer.

b) Let $M = (E, I)$ be a matroid. We denote by $\mathcal{B}(M)$ the set of bases of M. Now we define the concept of the **dual** of a matroid.
The dual $M^* = (E, I^*)$ of a matroid $M = (E, I)$ contains the same set E as the original matroid. The independent sets are formed as follows: $I^* = \{Z \subseteq E : \exists B \in \mathcal{B}(M) \text{ with } Z \cap B = \emptyset\}$.
(Note: The bases of a matroid are its maximal independent sets.)

   (i) Consider the matroid $M_1 = (E_1, I_1)$ with the set $E_1 = \{a, b, c, d\}$ and $I_1 = \{\emptyset, \{a\}\}$.
   State the dual's family of sets I_1^*.

   (ii) Show: The dual $M^* = (E, I^*)$ of a matroid M is an *independence system.*

Exercise S8. 2. *DGS Algorithm*

Consider the following valuation matrix, where bidders are considered to have *unit-demand* valuations, namely each bidder is interested in winning at most a single item in this auction.

	A	B	C
Bidder 1	5	8	3
Bidder 2	9	9	1
Bidder 3	6	2	4

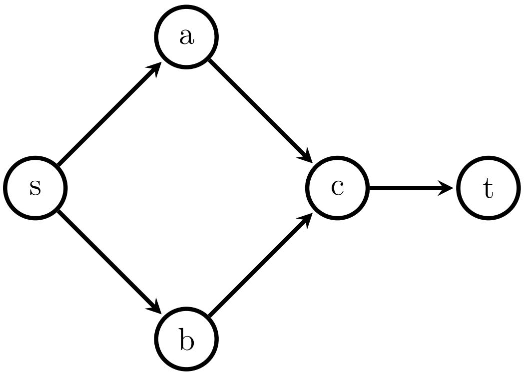
Compute the minimal competitive equilibrium prices and corresponding assignment using the DGS algorithm. Provide all intermediate steps.

Exercise S8. 3. *Maximum flow and minimum cut counterexamples*

Let $(G = (V, E), s, t, c))$ be a network with source s, sink t, and integral capacities $c(e) \in \mathbb{N}$ for all edges $e \in E$.

Consider the following statements. Use the given graphs to find a suitable counterexample for each one of them. Clearly describe why the given counterexample refutes the statement.

a) If all capacities are odd then there is a maximal $(s - t)$-flow f such that $f(e)$ is odd for all $e \in E$.



b) Adding a number $\lambda \in \mathbb{N}$ to all capacities $c(e)$ does not change the minimal cuts.